A Station-Level Air Quality Benchmark for Central Asia

Frozen leave-city-out splits, an operational-availability account, and a transfer baseline evaluated under whole-city holdout

Jaloliddin Musayev[1,*], **Asadbek Abdivayitov**[2], **Ozodbek Yo'ldashev**[3]

[1] International House Tashkent Academic Lyceum, Tashkent, Uzbekistan [2] First Specialized Boarding School, Karshi, Uzbekistan [3] National University of Uzbekistan, Tashkent, Uzbekistan

[*] Corresponding author: jaloliddin2009applicant@gmail.com ORCID iDs: Jaloliddin Musayev 0009-0003-0210-3687; Asadbek Abdivayitov 0009-0006-3484-3438.

Abstract

Central Asia carries some of the highest particulate burdens on earth and some of the sparsest monitoring, yet no open station-level benchmark exists against which competing estimation methods can be compared on identical terms. Regional results are typically reported under random cross-validation, which places observations from the same station on both sides of the split, answering a question about interpolation while appearing to answer one about unmonitored locations.

We present a reproducible benchmark over 7 instruments in 6 cities -- 5 reference-grade US-embassy monitors and 2 low-cost sensors in Khujand -- 2018-11-27 to 2024-12-31. Splits are frozen and checksummed before the reported results were produced: blocked-temporal with a purge gap of 240 hours from the maximum feature lag and horizon, plus leave-city-out over 6 folds. Nowcasting at unmonitored sites and forecasting at monitored stations are defined separately and never pooled. Ground observations are fused with satellite retrievals, chemistry-transport forecasts, reanalysis meteorology and static geography, each carrying a measured acquisition latency that governs admissibility.

Against a mandatory baseline ladder scored at a single temporal resolution, tuned gradient boosting reaches RMSE 28.01 ± 0.35 μg/m³ at unmonitored locations, the lowest of all 6 admissible baselines including inverse-distance weighting (29.44 μg/m³) — though that ordering is not statistically separable (paired p = 0.586) and reverses if one city is removed. The advantage over bias-corrected CAMS is **not statistically significant** once the city — the unit this protocol generalises over — is the unit of analysis (paired t on 6 city means, p = 0.1392; exact permutation p = 0.1250), and mean per-fold R^2 is -0.04 with 3 of 6 folds negative: the model ranks first on error while explaining little within-city day-to-day variation. Further results are reported against interest: no credential-free nowcaster beat a constant always-exceed classifier at a 61.8% base rate; no feature family dominates attribution, with satellite products (26.6%) ahead of calendar terms (22.2%), static geography (21.8%) and spatial neighbours (20.4%); and measured latency invalidated three of five initial availability assumptions. The contribution is the fixed evaluation protocol, and the shortcuts fixing it

forecloses.

Keywords: air quality; PM2.5; Central Asia; machine learning benchmark; spatial cross-validation; leave-city-out; satellite remote sensing; reproducibility

> **Reproduction.** Every figure below is regenerated by `make reproduce` (or `python tasks.py reproduce`) from the frozen splits in `benchmark/splits/splits.json`, checksum `544a044c…f9e0c6c`. No number in this document is typed by hand.

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1. Introduction

1.1 The gap

Central Asia is among the most polluted inhabited regions on earth. It is also among the least instrumented. Tursumbayeva et al. (2023) put annual PM2.5 in six regional capitals at 4.3–12.6 times the WHO 2021 guideline and trace the burden mainly to coal combustion rather than transport, contradicting the official emissions inventories. Source-apportionment studies across the region reach the same conclusion independently, in Kazakhstan (Tursun et al., 2025) and Tajikistan (Papagiannis et al., 2024). The monitoring base under those numbers is thin and unevenly open. Turkmenistan runs no national air quality network at all. Kazakhstan releases its data only to users physically inside the country. Only Kyrgyzstan publishes in a fully open form (OpenAQ, 2025).

Estimates are not what the region lacks. Global gridded products (van Donkelaar et al., 2021) already assign PM2.5 values across Central Asia, and the epidemiological literature consumes them. What nobody can do is check those estimates, or set two methods against each other on identical terms. There is no open station-level benchmark for the region. No frozen splits, no declared protocol, no shared evaluation.

The cost of that absence is measurable. Consider the closest environmental analogue in the literature: PM2.5 estimation across Xinjiang, an arid, dust-affected, sparsely monitored region much like this one. Jin et al. (2022) report R^2 between 0.73 and 0.81 under 10-fold cross-validation with no spatial stratification. With 41 stations in 16 cities and 8-day averaging, that design places observations from the same station, frequently the same window, on both sides of the split. The number answers one question. How well can we interpolate among stations we already have? Readers take it as answering a different one: how well can we estimate where no monitor exists? Nothing in a reported R^2 separates the two.

This is not a complaint about one paper. Roberts et al. (2016) show that spatially, temporally or hierarchically structured data require blocked validation, because ordinary k-fold leaves dependence straddling the split. Meyer et al. (2019) demonstrate the same for spatially derived predictors. Alazmi and Rakha (2022) measure the effect directly on particulate data, running random, spatial and spatiotemporal cross-validation side by side. Tang et al. (2024)

place validation strategy among the systematically overlooked issues in air quality machine learning. The methodological position taken here is therefore not new. What the region has never had is a benchmark that enforces it.

1.2 What this paper contributes

AQ-Bench (Betancourt et al., 2021) is the precedent: 5,577 stations worldwide, split by spatial clustering at a 50 km threshold. It targets long-term ozone metrics from station metadata, a time-independent regression, and it excludes Central Asia entirely. We borrow its spatial clustering rationale for leave-station-out instead of inventing a second one, and diverge on pollutant, target, temporal protocol and region. Section 3 states the differences precisely. AirDelhi (Chauhan et al., 2023) offers a second precedent for fine-grained particulate benchmarking, confined to a single city.

C1 — a benchmark. 7 stations across 6 cities. Splits were frozen and hashed before the results reported here were produced: blocked-temporal with a derived purge gap, leave-city-out over 6 folds, and leave-station-out where station density allows it (2 folds; Almaty, Ashgabat, Bishkek, Dushanbe, Tashkent hold one instrument each and are named ineligible, not quietly dropped). A test enforces immutability. It fails for the authors exactly as it fails for anyone else.

C2 — transfer evaluated without flattery. We offer no new method. We measure how far spatial transfer survives into an aerosol regime unlike the source domain, under a protocol the region has never applied. That framing makes a negative result publishable. Several of ours are negative.

One fold deserves naming here. Khujand's stations begin after the training block closes, so the city contributes no training label anywhere in the record. Not one row. The model arrives with no local history of any kind and has to return a concentration regardless. This is the harshest test the benchmark contains, and it is also the one that matches the deployment case the work exists for: an unmonitored city asking for a number it has never been given. A model that collapses on Khujand has not earned the right to be deployed anywhere new. We report the fold on its own. Averaging it into the other five would report neither.

Estimating surface PM2.5 from satellite columns is itself an established line of work (Zang et al., 2017; Xu et al., 2018); what is new here is subjecting it to a spatial protocol that withholds whole cities.

C3 — an operational-constraint account. Every predictor carries a measured latency and a typed availability flag. Anything that cannot exist at prediction time is barred from deployable configurations by test rather than by convention.

1.3 Findings that shaped the work

Three results shaped the work that follows.

A constant is hard to beat, and that says more about the region than about the models. No credential-free nowcaster beat a trivial always-exceed predictor on health-relevant exceedance. The reason is uncomfortable and entirely physical: PM2.5 in these cities clears the WHO 24-hour guideline on most days of the year, so a classifier that never varies is correct most of the time. Chronic pollution, not modelling skill, sets that floor. Beating it is the only evidence that a model has learned something past the regional mean. Raw CAMS, a full chemistry-transport model, is improved simply by subtracting a per-city constant, and even after that correction it reaches only R^2 -0.11. Across two phases, every baseline we tried sat below the accuracy of predicting the held-out city's own mean.

Metric choice reverses conclusions. RMSE and exceedance skill rank the same models in nearly opposite orders. Smoothing toward the mean lowers squared error while destroying the variance needed to tell a bad day from a good one, so a model wins one metric by losing the other. Report either alone and the ladder comes out backwards for the other task.

The headline modelling result is mixed, and both halves are reported. Tuned gradient boosting with spatial neighbour encodings reaches RMSE 28.01 ± 0.35 μg/m³ at unmonitored locations, ahead of every admissible baseline — including inverse-distance weighting at 29.44 μg/m³, a margin of 1.43 μg/m³. That is a real result on error.

It is not a result on skill. Mean per-fold R^2 is -0.04, ranging -0.55 to 0.52, with 3 of 6 folds negative — in those cities the model does worse than a flat line through the city's own mean. Ranking first on RMSE while explaining little within-city variation is not a contradiction: most of the achievable error reduction in this region is in getting a city's *level* right, which is precisely what a benchmark built on whole-city holdout is designed to expose.

Its comparison against bias-corrected CAMS (29.77 μg/m³) is not significant under the unit of generalisation this protocol is built on: a paired test over 6 city means gives p = 0.1392, and an exact permutation test 0.1250. Per-fold Diebold–Mariano tests do clear 0.05 individually, but six tests are run and **only 3 of 6 survive Holm correction**. We report the corrected count, not the raw one.

1.4 What this paper does not claim

We do not claim state of the art. No like-for-like comparison exists on these splits, for the plain reason that we are the ones creating them. Nor do we claim to introduce transfer learning for data-poor regions. Gupta et al. (2024) hold that ground already, proposing a latent dependency factor for spatial transfer of PM2.5 estimation and reporting a 19.34% gain over baselines across ten target sensors against eastern-US source data. Theirs is a method paper, not a benchmark, and it defines no reusable public split. What remains available to us is the evaluation protocol and the region, not the idea. Related transfer approaches have since been reported for other data-sparse settings, including African cities (Mazuruse et al., 2026) and hybrid deep architectures (Ni et al., 2022). We also decline to compare our figures against published R^2 values obtained under random cross-validation. A leave-city-out number and a random-CV number are not commensurable, and treating them as though they were is the precise error this benchmark exists to prevent.

One piece of prior art we cannot resolve. A 2025 conference abstract claims the first machine-learning PM2.5 prediction for Tashkent, using ten automated stations with weather and seasonal inputs. We could not obtain it. The publisher returns HTTP 403, and the record appears in none of OpenAlex, Crossref, Semantic Scholar or Europe PMC. **Its split protocol is unverified.** We therefore make no claim about what that work did or did not do, and we specifically do not assert priority for applying leave-city-out to Tashkent. Should it prove to be spatially stratified, the novelty of C1 narrows to the multi-city benchmark, not the city. Section 7 carries this as an open item.

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2. Data, Operational Constraints, and Informative Missingness

2.1 Ground truth

PM2.5 comes from 7 instruments across 6 cities: Almaty, Ashgabat, Bishkek, Dushanbe, Khujand and Tashkent. **5 are US diplomatic-post reference monitors and 2 are Clarity low-cost sensors (Khujand)** — see Section 7.2. The embassy monitors are the only consistent multi-country reference in the region and the sole route to any measurement in Turkmenistan.

That network contracted sharply in March 2025, when the StateAir publication channel closed. Five of the ten contributing source feeds stop on 2025-03-04, and after co-published feeds are merged **2 of 7 benchmark stations (8881, Bishkek) end there**; the rest survive through a longer-lived feed. Reporting states that seventeen years of archive were subsequently removed from the originating platform.

The closure was not uniform, and an earlier claim here that the programme had simply ended is withdrawn. A live query on 2026-08-14 found three of the same diplomatic-post monitors still republished through the AirNow provider after the StateAir channel closed — Ashgabat to 2025-09-24, Almaty to 2025-11-14, and Dushanbe still reporting — with measurements retrieved and verified rather than inferred from metadata. The record this benchmark curates is nevertheless fixed: the panel is frozen, the test year is 2024, and extending it would change the benchmark and require a version bump rather than a refresh.

Quality control was pre-registered before the data were inspected, so no rule could be tuned to improve a count. Seven rules cover physical range, flatlining, zero-runs, unit sanity, duplicate identity, timezone correctness and completeness. Two produced findings worth reporting.

Duplicate identity, and the case distance could not see. The embassy monitors are published twice — once by StateAir, once by AirNow — under distinct identifiers 57 m apart in Bishkek and 40 m in Ashgabat. Exact coordinate matching does not catch this. Under leave-station-out the held-out station would have been the same physical device as one in training. Detection is therefore distance-based at 150 m.

A distance rule cannot catch the inverse case, and this benchmark contained one. Dushanbe was published by the same two programmes, but its two records carry coordinates **6.06 km apart**, so the 150 m rule passed them and an earlier version of this manuscript cited that separation as evidence the sites were distinct. They are not. Over the 33,462 hours in which both report, **94.0% of readings are bit-identical**, and of the remainder 99.9% match exactly at a five-hour offset — Dushanbe is UTC+5, so those are the same measurements timestamped in local time rather than UTC. **99.99% of overlapping hours are the same reading.** The comparable Khujand pair, 14.4 km apart, is bit-identical on 0.3% of hours.

A second rule (Q5c) was therefore added: flag any station pair whose overlapping observations are bit-identical on more than half of samples, regardless of separation. Two independent instruments do not agree to floating point. The Dushanbe records are merged under the same precedence-and-gap-fill rule already applied to Bishkek and Ashgabat, giving 7 instruments rather than eight, and the benchmark is versioned **1.1.0** to mark that the split changed. Sections 6 and 7 report how the modelling results moved as a result; they moved against the model.

Timezone correctness. Metadata offsets are not trusted. Each station's diurnal composite is cross-correlated against a regional reference, and an initial implementation rejected both Khujand sensors for an apparent 12-hour shift. Investigation showed the regional reference self-correlates at r = +0.71 under a 12-hour rotation, because Central Asian urban PM2.5 is bimodal with peaks roughly half a day apart. Whole-shape correlation cannot separate a half-day offset from none in such a signal. The check now reports lag identifiability and flags rather than rejects when the hypothesis is not distinguishable — a station was nearly deleted by a discriminator reporting an artefact.

Figure 1 shows the station geography; marker area is proportional to the number of retained hourly observations.

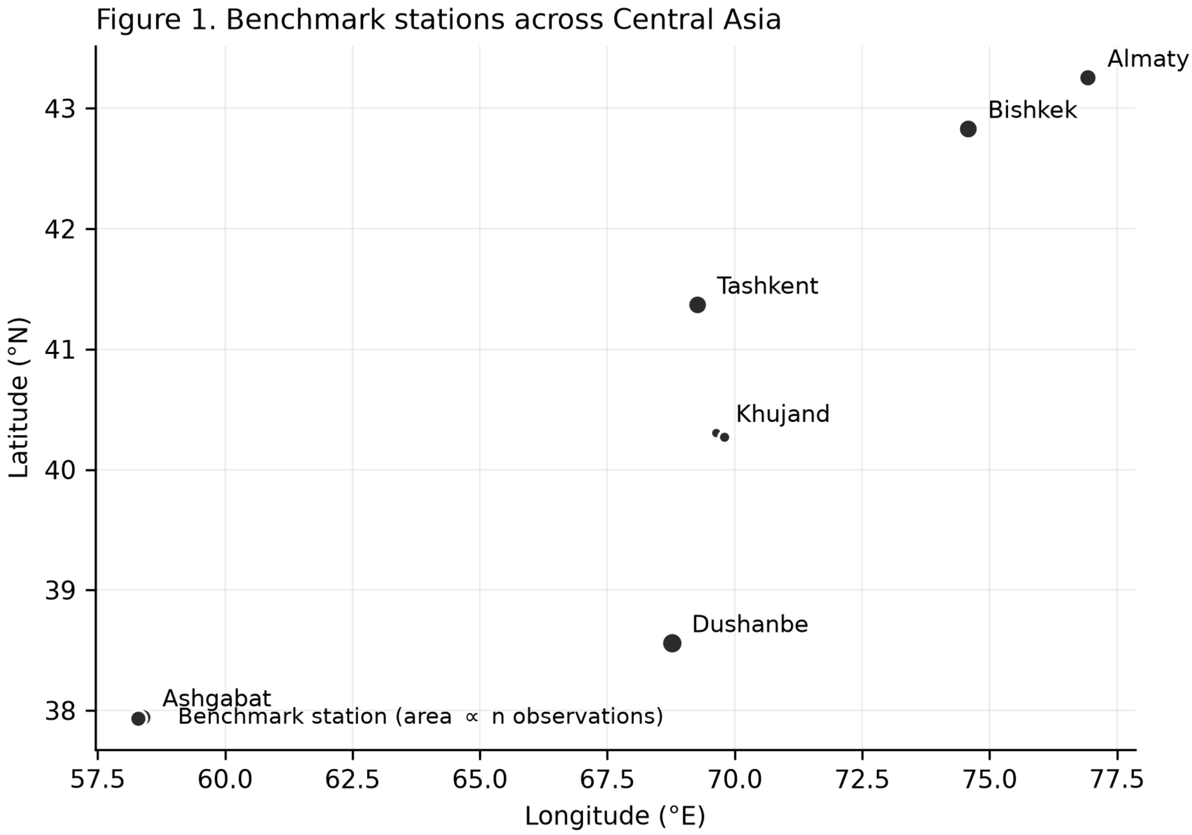

Figure 1. Benchmark stations across Central Asia. Marker area is proportional to the number of retained hourly observations. Six cities span five countries; Turkmenistan (Ashgabat) and Kazakhstan (Almaty) contribute one station each.

2.2 The splits

Train 2018-11-27 to 2022-12-31; validation 2023-01-11 to 2023-12-21; test 2024-01-01 to 2024-12-31. Purge gaps of 240 hours separate the blocks.

The purge is derived, not chosen. A training sample at t reads features over [t–max_lag, t] and predicts t+h, so the gap must satisfy purge ≥ max_lag + max_horizon = 168 + 72 = 240 hours. A test recomputes this from the model definitions, so introducing a model with a longer feature window fails the build rather than silently leaking across the boundary.

Test year 2024 is forced by the embassy shutdown. It is the last complete year with reference-grade coverage. A post-shutdown block would cover two cities, not six.

Khujand contributes no training rows at all. Both its stations begin in late 2023, after the training block closes, so Khujand appears only in validation and test. This passed the completeness rule because that rule checks span and coverage, not overlap with the training block. We retain it deliberately: it constitutes a pure zero-shot spatial transfer fold, in which the model has no local history in any form. Its fold is strictly harder than the others and is reported separately rather than averaged in.

2.3 Operational availability as a typed property

Reanalysis products describe exactly the quantities that drive PM2.5, are better documented than their forecast counterparts, and already sit in most atmospheric pipelines. A model consuming ERA5 boundary-layer height to predict tomorrow's air quality will post an excellent number and cannot be deployed. Nothing in its RMSE looks wrong.

Availability is therefore declared per feature and enforced by test. Every measured latency in this study is given in Table 2.1; **three of five initial estimates were wrong, one by more than three orders of magnitude.**

Product	Claimed	Measured	Status
MAIAC AOD (Earth Engine, MCD19A2.061)	6 h	~8 days	oracle only
Sentinel-5P AAI (OFFL)	5 h	~72 h	oracle only
Sentinel-5P AAI (NRTI)	—	< 24 h	deployable
CAMS forecast PM2.5	12 h	~10 h	**deployable**
ERA5 reanalysis	~5 days	163 h	oracle only
VIIRS active fire (v001)	4 h	**774 days**	superseded

The VIIRS case is the instructive one, and the latency was the lesser problem. The originally-mapped collection is deprecated and its final asset falls on 2024-06-16 — dead centre of the frozen test year, giving 161 images in January–June and **zero** in July–December. A model using it would have seen fire signal for half the test period and structurally none for the other half, producing an apparent regime change on 1 July that invites a meteorological explanation for a dead data feed. A latency check catches the 774 days; only a coverage check against the *frozen* test block catches the mid-block termination. Both are now enforced.

CAMS products derive from the Copernicus Atmosphere Monitoring Service assimilation and forecast system (Inness et al., 2019). CAMS requires a second operational distinction: forecast step zero has assimilated observations at the valid time, so using it to predict that time is lookahead wearing a forecast label. All CAMS features use the 24-hour lead, which is what a live service holds.

2.4 The R7 phenomenon: missingness correlated with the target

Two aerosol regimes operate here and fail retrieval differently: winter coal combustion, and spring salt-dust transport from the desiccated Aral bed (Banks et al., 2022; Liu et al., 2025).

Satellite retrieval does not fail at random. It fails during dust, cloud and snow — the conditions that accompany the highest concentrations. Dropping incomplete rows therefore conditions the evaluation on *"retrieval succeeded"* and biases every result toward calm, clear, low-concentration days.

We quantified this across all five satellite features on daily station-matched data. Each satellite product is extracted onto a complete station × date grid — the compositor emits a fully-masked image on days with no usable granule rather than omitting the row — so a failed retrieval is a masked cell, not an absent one, and the grid is the denominator. Rates below are conditional on the station-day also carrying a ground observation, since that is the subset on which the association can be tested at all.

Feature	Retrieval	∆ median PM2.5 on missing days	*p*	Retrieval on worst decile
Sentinel-5P SO_2	57.4%	**+10.3**	1.6e-138	**14.0%**
MAIAC AOD	62.9%	+5.6	1.4e-44	38.1%
Sentinel-5P NO_2	70.7%	+3.3	1.7e-14	58.0%
Sentinel-5P CO	81.1%	+2.6	7.2e-05	81.9%
Sentinel-5P AAI	99.8%	+6.8	0.75	99.9%

SO_2 is blind in the season it exists to observe. It retrieves on 0.1% of December days against 91.0% in July. Retrieval requires ultraviolet signal, and at 38–43°N in December the solar zenith angle leaves too little; this is a geometric floor, not a cloud accident. The direct tracer for the region's dominant winter source is unavailable throughout that source's season. Additionally, 29.8% of the retrievals that do occur are negative, sitting below the noise floor — clipping them at zero would bias the coal tracer upward across a third of its observations.

MAIAC (Lyapustin et al., 2018) has been validated specifically over Central Asia (Chen et al., 2021), which is why it was selected over coarser aerosol products. Its bright-surface retrieval matters here: much of the study region is high-albedo desert and seasonal snow, the conditions under which coarser algorithms lose the surface-aerosol separation.

The split between clean and contaminated features follows retrieval physics. CO uses the 2.3 µm shortwave-infrared band and AAI uses ultraviolet reflectance *ratios* rather than absorption depth (Veefkind et al., 2012); both survive winter geometry and cloud, and neither shows target-correlated missingness. MAIAC (visible/near-infrared) and the ultraviolet absorption retrievals do not.

The two clean features are complementary to the contaminated ones. AOD and AAI are both present on 62.8% of station-days; AAI alone covers a further 36.9%; neither is available on 0.1%. Usable satellite coverage therefore rises from 62.8% to 99.9%, and the coverage AAI adds is concentrated precisely where MAIAC fails.

Consequently missingness is **modelled, never dropped**. Valid-pixel counts are promoted to features in their own right. They are retained for Task F but **excluded from Task N** — see

Section 5.3, where an ablation showed retrieval-count features to be city-specific rather than transferable. Section 6 reports attribution for the features actually used, so the retrieval-count features carry no Task N attribution at all: they were excluded from it. The evidence that *when* a retrieval failed carries information is their retention in Task F and the Section 5.3 ablation, not an attribution share.

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3. Benchmark Construction

The benchmark is the primary contribution. This section states what is frozen, what a submission must satisfy, and which failure modes are enforced mechanically rather than requested in prose.

3.1 What is frozen

`benchmark/splits/splits.json` fixes 7 stations across 6 cities, the three temporal blocks, 6 leave-city-out folds and 2 leave-station-out folds. It is accompanied by `splits.sha256`. The checksum is verified in CI and by `make reproduce`; any edit to the split file fails the build.

The split design follows the blocked-validation position of Roberts et al. (2016) for data with spatial and temporal structure, and adopts the spatial-clustering rationale of AQ-Bench (Betancourt et al., 2021) for leave-station-out. Leave-one-out protocols carry their own failure mode — Austin et al. (2025) show that distributional bias can compromise them — which is why leave-*city*-out, not leave-station-out, is the headline protocol: it holds out a whole aerosol regime rather than one instrument within a regime.

The checksum is over bytes, which on a mixed-platform repository is not automatic. An early version was unverifiable because `Path.write_text` applied CRLF translation on Windows, and the `.gitattributes` fix that followed was itself wrong: for attributes the **last** matching line wins, the opposite of `.gitignore`. The general `* text=auto` rule must therefore be declared *before* the `benchmark/splits/** -text` override, not after. This was caught only by cloning into a temporary directory and running `sha256sum -c` there — an in-place check passes on a file that no other machine can reproduce.

3.2 Two tasks, never pooled

	Task N — nowcasting	Task F — forecasting
Question	concentration at an **unmonitored** location, now	concentration at a **monitored** station, later
Protocol	leave-city-out	blocked temporal, t+24 h, t+48 h, t+72 h
Target history	**inadmissible**	admissible

Spatial features	admissible (neighbours exclude the held-out city)	admissible

Under leave-city-out the held-out city contributes no label, so a local autoregressive lag is undefined at inference — not merely optimistic, but unavailable. Under blocked-temporal forecasting the same lag is exactly what a deployed service holds. A single table containing both tasks would compare models with access to the target's own history against models without it, and the resulting ranking would be an artefact of that asymmetry. The two are reported separately throughout, and a test asserts that no results file mixes them.

3.3 The purge gap is derived

Purge = max_lag + max_horizon = 168 + 72 = 240 hours, applied at both block boundaries. A test recomputes this from the registered model definitions rather than reading a constant, so adding a model with a longer feature window fails the build instead of quietly leaking across the boundary. The failure mode this prevents is specific: a 168-hour rolling feature computed at the first test timestamp reads 168 hours of training labels.

3.4 Feature admissibility is typed, not documented

Every feature carries `available_at_runtime` and `latency_hours` as properties, and feature sets are constructed by filtering on them:

- `static_only` — geography and land surface; no temporal input at all
- `deployable` — everything a live service can hold at inference time
- `benchmark_retrospective` — adds products whose latency exceeds the forecast horizon
- `reanalysis_oracle` — deliberately inadmissible; exists to quantify the gap

The oracle set is retained precisely because it is not deployable. Reporting it alongside the deployable set converts an invisible methodological error into a measured quantity: the difference between the two is the cost of the operational constraint, and Section 6 reports it. A guard test constructs a contaminated feature set and asserts the admissibility check rejects it, so the check cannot silently degrade into a no-op.

3.5 Khujand: zero-shot spatial transfer

Both Khujand stations begin after the training block closes, so Khujand appears only in validation and test. Its leave-city-out fold is therefore strictly harder than the other five: the model has no local history in any form, not merely no label in the current fold.

We retain it as a distinct evaluation regime rather than repairing it. Every other fold measures interpolation between cities the model has seen at some point in training; Khujand measures extrapolation to a city that never appears. Averaging the two would report neither. Khujand is reported both inside the pooled figure and separately, and its fold is labelled zero-shot wherever it appears.

3.6 What a submission must report

1. Both tasks, in separate tables, with the ladder of Section 4 included.
2. Mean ± standard deviation over at least five seeds. Deterministic models report `(det.)` rather than `± 0.000`, so that a zero is never mistaken for a measured variance.
3. Per-city results, not only the pooled figure. Six cities with different aerosol regimes averaged into one number hides the variation that matters.
4. Diebold–Mariano tests (Diebold and Mariano, 1995) for any claimed improvement, with the truncation lag stated, Newey–West HAC variance (Newey and West, 1987) and the Harvey–Leybourne–Newbold small-sample correction (Harvey et al., 1997). Section 4.4 shows why the lag cannot be left at its default: setting it to zero inflated our *p*-values by seven orders of magnitude.
5. Exceedance metrics accompanied by the trivial-classifier score and Peirce skill (Section 4.3), because F1 alone does not establish skill.
6. A statement of which feature set was used, from the four above.

3.7 Reproduction

`make reproduce` runs the pipeline end to end: lint, type check, the full test suite, checksum verification, split regeneration, the baseline ladder, **the model layer** (untuned and tuned gradient boosting, the CAMS variants, and the significance analysis), table regeneration and manuscript rendering — in that order, because splits are hash-verified before any model reads data. Every number in this paper is produced by that command. No figure in the manuscript is typed by hand; Section 7.5 describes the mechanism and the drift it caught. The command is idempotent: a second run reproduces all 29 result tables **byte-identically**, verified by SHA-256.

A reproducibility failure this paper had to fix in itself. Until benchmark v1.1.0 the model layer was *absent* from `make reproduce`. Its four producers wrote files under names that were then renamed by hand into the names the manuscript cites, so the command regenerated only the Section 3 baseline tables and still exited 0 — while every headline table sat unchanged on disk. The claim "every number is produced by that command" was therefore false when first published, not because any number was fabricated (all of them regenerate exactly) but because the chain that was supposed to guarantee it had a gap. Each producer now writes the name the paper cites, the rename step is gone, and a test asserts that every table read by the manuscript has a producer inside the reproduction chain. We report this because a reproducibility claim that has never been executed end to end is exactly the kind of claim this benchmark exists to make checkable.

On the ordering of the freeze. The splits carry a signed timestamp and are immutable by test. What the repository can prove is that **the modelling results reported here were**

produced after the splits were frozen and hash-verified. It cannot prove the stronger claim, made in earlier drafts, that no model had been fitted on any data before the freeze: model *code* is committed roughly twelve hours before the freeze timestamp, no result artefact is under version control from that period, and the split builder was committed in the same commit as the splits. We therefore state the weaker claim, which the evidence supports, and flag the stronger one as unverifiable. A future benchmark version should record run manifests with content hashes so the ordering is provable rather than asserted.

What that command does and does not require. The frozen splits are committed, so *verifying* the benchmark needs neither credentials nor a rebuild — `sha256sum -c splits.sha256` is sufficient, and the test suite runs offline against committed fixtures. *Regenerating* the numbers additionally requires the derived ground-truth panel, which is built from the OpenAQ archive with an API key. That panel is not redistributed here because the per-station licence terms are not yet transcribed; `data/MANIFEST.md` records the provenance and `README.md` the two commands that rebuild it. Running the pipeline without it fails with those instructions rather than a missing-file traceback. We state this explicitly because "one command reproduces everything" is a claim reviewers test, and it is true only after that acquisition step.

Two properties of that command are load-bearing and were not true of earlier versions. First, it **regenerates the manuscript, not only the tables**: an earlier `paper` target stopped after building the CSVs, so the rendered sections were never rebuilt from the figures `reproduce` had just recomputed, and the zero-drift guarantee held only halfway. Second, a missing tool is a **failure, not a skip** — the task runner once printed `SKIP (not installed)` and continued, so on a machine without `ruff` the command exited 0 having verified nothing. A reproduction harness that passes without doing the work is worse than no harness, because it converts an unrun check into a green light.

Environment setup is `python scripts/setup_env.py`, which works with or without `uv` and locates a Python 3.12 by *executing* candidate interpreters rather than trusting a registry. On our own Windows machine the `py` launcher advertised a 3.12 runtime that failed to start. Where `make` is unavailable, `python tasks.py <target>` runs the identical commands — the Makefile delegates to the same table, so the two cannot diverge.

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4. The Baseline Ladder

A learned model is only interesting relative to what it beats. This section reports the mandatory ladder — persistence, climatology, raw chemistry-transport, and the spatial interpolators — before any learned model appears. Every rung is credential-free and deterministic, so seed variance is zero by construction and is reported as `(det.)` rather than as a measured `± 0.000`.

4.1 Task F — forecasting at monitored stations

Test block 2024, horizons t+24 h, t+48 h, t+72 h, pooled across stations.

Model	RMSE (µg/m³)	R^2
persistence, y(t)	40.14	-0.34
diurnal persistence, y(t–24 h)	40.14	-0.34
climatology (station mean)	36.24	-0.15
same-hour 7-day mean	33.23	0.11

Persistence and diurnal persistence are numerically identical, and this is correct. All three evaluated horizons are multiples of 24 h, so "the same hour yesterday" and "lag-24 persistence" resolve to the same timestamp. The Diebold–Mariano procedure reports them as exact ties — the loss differential is identically zero and the test statistic is undefined rather than insignificant. We report the degeneracy rather than dropping a rung, because a ladder in which two rungs coincide at the evaluated horizons is a property of the horizon grid that a reader needs in order to interpret the table.

Degradation with lead time is visible only in the persistence family (37.76 → 41.57 µg/m³ from t+24 h to t+72 h). Climatology is flat by construction at 36.24 µg/m³, since it does not read the recent past. **Only the same-hour 7-day mean achieves positive R^2** (0.11); every other rung is worse than predicting the test-block mean.

Figure 2 shows the full Task N ladder, including the tuned model of Section 5 for context. Every interpolation rung sits above the training pool mean.

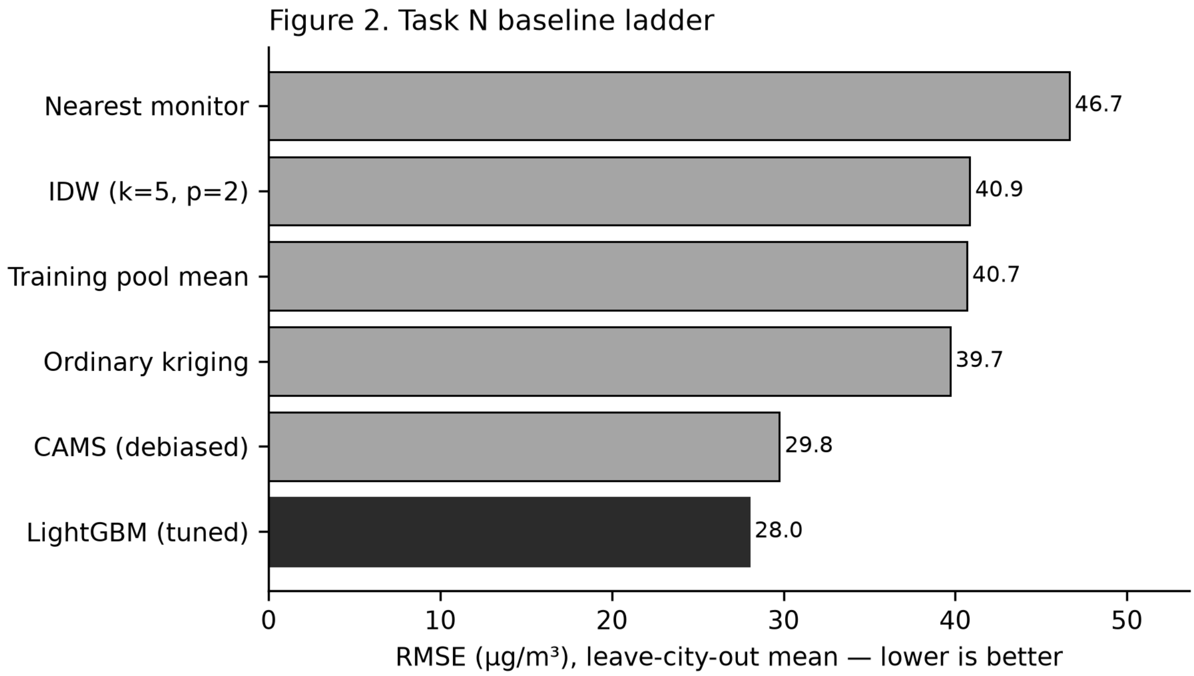

Figure 2. Task N baseline ladder, leave-city-out mean RMSE (µg/m³). Lower is better. The tuned model of Section 5 is shown in dark fill for context; every credential-free interpolation

rung sits above the training pool mean.

4.2 Task N — nowcasting under leave-city-out

The held-out city contributes no label. These are pure spatial interpolators, fitted on the remaining five cities.

Model	RMSE (µg/m³)	R^2	Exceedance F1	Peirce skill
nearest monitor	46.66	-0.79	0.753	0.436
IDW (k=5, p=2)	40.86	-0.26	0.775	0.378
training-pool mean	40.69	-0.24	0.741	0.000
ordinary kriging	39.71	-0.22	0.736	0.296

Every spatial baseline has negative R^2. Interpolating between Central Asian cities hundreds of kilometres apart is worse than predicting a constant. That is the floor the task sits on, and it is why the leave-city-out result in Section 6 is modest.

The training-pool mean is an explicit rung of the ladder. It is a constant: the mean of all training-block labels. It was promoted to the ladder after we observed that it outranks two of the three genuine interpolators on RMSE (40.69 vs 40.86 and 46.66). Any model that does not clear a constant has not demonstrated spatial skill, and without this rung in the table the reader cannot check that.

Kriging attains the best RMSE (39.71) and the **worst** exceedance F1 (0.736). Optimising squared error pulls predictions toward the mean, which suppresses exactly the excursions the exceedance metric scores. Reporting one number without the other would support two opposite conclusions about the same model. We additionally report the kriging fallback rate: where the system is singular the implementation silently degrades to IDW, and a column labelled "kriging" that is substantially IDW underneath is a reporting error rather than a modelling one.

4.3 Exceedance F1 does not measure skill

The WHO 2021 24-hour guideline is 15 µg/m³, scored on local-calendar daily means because that is what the guideline defines. In this region the exceedance base rate is **61.8%** — exceedance is the common case, not the rare one.

A classifier that predicts "exceeds" unconditionally therefore scores F1 = **0.741**. The training-pool mean — a constant, carrying no information whatsoever — scores 0.741, and is the **highest-F1 model in the entire Task N ladder**. Its Peirce skill is 0.000, exactly zero, as it must be for any constant.

The consequence reaches past this table: a paper reporting only exceedance F1 on this region could present a constant as its best classifier and the number would look respectable. We therefore report, alongside every exceedance F1:

- `f1_trivial_always` — the score of the unconditional classifier on the same rows
- `beats_trivial` — whether the model exceeds it at all
- `base_rate` — so the reader can compute the trivial score independently
- `peirce_skill` — TPR – FPR, which is base-rate independent and **zero for any constant**

Peirce skill separates the ladder in the way F1 does not: nearest monitor (0.436) and IDW (0.378) carry genuine discriminative information, while the constant carries none, and the F1 column ranks them in the opposite order.

4.4 CAMS as a baseline

CAMS is the operational chemistry-transport forecast for the region and the strongest credential-free comparator. It is used at the 24-hour lead throughout: forecast step zero has assimilated observations at the valid time, so scoring it against that time is lookahead wearing a forecast label.

Variant	RMSE (μg/m³)	R^2	Bias (μg/m³)
raw	35.27	-0.39	-21.34
locally debiased	30.35	-0.18	-2.71
pooled debiased	30.68	-0.11	-2.71

Raw CAMS under-predicts by -21.34 μg/m³ — a large systematic deficit consistent with an emissions inventory that does not capture residential coal and biomass combustion at Central Asian intensity. Removing that bias improves RMSE substantially (35.27 → 30.35) while R^2 remains negative (-0.18): the correction fixes the level, not the timing.

The two debiasing variants are distinguished by protocol. **Local debiasing is admissible only in Task F**, since it fits a per-city offset on that city's training labels. Under leave-city-out no such labels exist, so Task N uses the pooled correction fitted with the held-out city excluded. The pooled variant is the weaker of the two (30.68 vs 30.35), and reporting the local figure under a leave-city-out heading would overstate the baseline the learned model must beat — making the model's margin in Section 6 look smaller than it is, in this direction, but the protocol violation is the same either way. A test asserts that a bias fitted on the full record differs from one fitted on the training block, so that a silently-leaking implementation cannot pass.

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5. Models and Training Protocol

5.1 Model class

Gradient-boosted decision trees (LightGBM) are the learned model. The choice is deliberate and is not a claim that trees are the strongest possible architecture. Three properties matter for this benchmark:

1. **Native missing-value handling.** Section 2 established that satellite retrieval failure is correlated with the target. A model that requires imputation forces a choice between discarding the informative rows and inventing values for them; LightGBM learns a default direction per split, so missingness is used rather than filled.
2. **Sample efficiency.** Six cities is a small spatial sample. A deep architecture would be tuned on validation folds of a few hundred station-days, and the tuning variance would exceed the effect being measured.
3. **Attribution that survives inspection.** Section 6 depends on being able to say which feature families carry the model, and to show that the answer is uncomfortable.

Reporting a deep model here would require a like-for-like comparison we cannot yet support at this sample size; the benchmark exists so that such a comparison can be made later on fixed splits.

5.2 Feature families

- **spatial neighbour** — inverse-distance-weighted neighbour concentration, neighbour mean, neighbour count. Under leave-city-out the neighbour set **excludes the held-out city**, which is enforced by passing the held-out label into the constructor rather than by filtering afterwards.
- **static geography** — elevation, terrain basin indices at 6-city scale, population density, distance to the Aralkum.
- **calendar** — cyclical day-of-year and hour encodings, weekday.
- **satellite** — MAIAC AOD, S5P AAI/NO_2/SO_2/CO.
- **satellite missingness** — valid-pixel counts and retrieval indicators, promoted to features in their own right.
- **CAMS forecast** — the 24-hour-lead chemistry-transport prediction.

Autoregressive lags of the target are **Task F only**. They are not merely optimistic under leave-city-out; they are undefined, because the held-out city supplies no history. For Task N the spatial features above are the only route to target information, and they are constructed with the held-out city removed.

5.3 Tuning

Hyperparameters are selected on the 2023-01-11 to 2023-12-21 validation block and then frozen. The test block 2024-01-01 to 2024-12-31 is read exactly once per reported configuration. The tuning protocol follows established guidance for tree ensembles (Probst et al., 2019). No hyperparameter, feature-set choice, or early-stopping decision is made against test-block performance.

The untuned and tuned configurations are reported side by side below. **They differ in four respects, not one, and the gap between them must not be read as the effect of hyperparameter tuning alone.** An earlier version of this manuscript described them as sharing "the same feature set". That was incorrect. The differences are:

	Untuned (`train_gbdt.py`)	Tuned (`train_phase5.py`)
features	tier columns only	tier columns **+ 3 spatial neighbour features**
trees	600	800
training rows	train block only (to 2022-12-31)	train block **+ validation block** (to 2023-12-21)
hyperparameters	library defaults	grid search, 16 combinations

Because the feature set, the tree count and the training window all change together, the untuned-to-tuned difference is a **combined** effect. This paper does not run the ablation that would separate them, and no causal attribution to tuning is claimed.

Feature set	Untuned RMSE	Tuned RMSE	Untuned R^2	Tuned R^2
static only	32.19	28.63	-0.73	-0.15
deployable	29.25	28.56	-0.21	-0.09
retrospective	29.51	28.01	-0.25	-0.04

The untuned column is retained because it was once used to argue that a poor result would have been an artefact of library defaults. On benchmark v1.1.0 that argument no longer holds: the tuned configuration now leads every admissible baseline (Section 6.1), while both configurations explain little within-city day-to-day variation. The combined feature-plus-window-plus-hyperparameter change improves RMSE, and RMSE leadership is not by itself evidence of skill.

5.4 Seeds and variance

Every configuration is run with five seeds and reported as mean ± standard deviation. Seed variance is small: the maximum across configurations is 0.25 µg/m³ for Task F and 1.79 µg/m³ for Task N.

Fold-to-fold variance is an order of magnitude larger than seed variance. The standard deviation of Task N RMSE across the 6 leave-city-out folds is 11.38 µg/m³, against 1.79 µg/m³ across seeds. Which city is held out dominates which seed was drawn. A study reporting seed error bars alone would communicate a precision this benchmark does not have, and per-city results are therefore mandatory (Section 3.6).

5.5 Operational cost of deployability

The deployable feature set excludes every product whose latency exceeds the forecast horizon; the retrospective set adds them back. The difference is the price of being deployable:

Task	Deployable	Retrospective	Cost
N (leave-city-out)	28.56	28.01	small
F (forecasting)	21.58	21.54	small

The cost is negligible in both tasks — 28.56 against 28.01 µg/m³ under leave-city-out, well inside the fold-to-fold spread of 11.38 µg/m³. This is a positive result for deployment and a deflationary one for the satellite features: products that arrive too late to be operational are also contributing little when they are available. Section 6.4 shows why.

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6. Results

All figures are on the frozen test block 2024-01-01 to 2024-12-31, five seeds, with Task N and Task F reported separately throughout.

6.1 Task N — leave-city-out nowcasting

Model	RMSE (µg/m³)	MAE	R^2
best spatial baseline (kriging)	39.71	—	-0.22
training-pool mean (constant)	40.69	—	-0.24
CAMS, pooled debias	29.77	21.19	-0.14
LightGBM, static only	28.63 ± 0.17	18.39	-0.15
LightGBM, deployable	28.56 ± 0.52	17.72	-0.09
LightGBM, retrospective	28.01 ± 0.35	17.81	-0.04

Learned-model RMSE is given as **mean ± standard deviation over 5 seeds**, where the resampled quantity is the whole leave-city-out protocol per seed. Baseline rungs are deterministic and carry no seed dispersion. Section 3.6 rule 5 requires this of every submission to the benchmark; it is applied here to the reference implementation as well.

Two numbers describe this result, and they point in opposite directions.

Between cities, the model retains some signal. Pooled over all evaluation rows — variance measured against the global mean — it explains **R^2 = 0.13**. Part of the contrast between a Dushanbe and a Bishkek is learnable from geography, satellite retrievals and neighbouring monitors.

Within a city, it is inconsistent. The headline **R^2 = -0.04** is the **mean of the 6 per-fold R^2 values**, each computed against *that city's own* mean. Scored this way the model must explain day-to-day variation inside a city it has never seen. The spread is wide — **-0.55 to 0.52**, with 3 of 6 folds negative (Section 6.1c). It succeeds in some held-out cities and does worse than a flat line through the city's own mean in others, and the average of those outcomes is near zero.

The two statistics are not in conflict; they answer different questions, and only the second is the task this benchmark defines. The correct summary is that **the model captures some between-city variation and within-city skill that does not generalise across cities.** A reader given only the pooled figure would substantially overestimate what it does.

The learned model has the lowest RMSE of any admissible method, and that ranking is not robust. Both halves matter and both are reported.

Scored on identical rows at identical resolution it reaches 28.01 ± 0.35 µg/m³ against 29.44 µg/m³ for the strongest legal rung (`idw_k5_p2`), a margin of 1.43 µg/m³, and it leads all 6 legal rungs on the fold mean.

But a fold mean is an average over 6 numbers, and the per-city differences are much larger than the average of them. Against inverse-distance weighting the model is better in 4 of 6 cities, with per-fold differences ranging from –10.73 to +5.10 µg/m³. A paired test over the 6 cities gives ***p* = 0.586** — the margin is **not statistically distinguishable from zero** at the unit of generalisation this benchmark is built on. Removing any single city and recomputing, the model still leads inverse-distance weighting in 5 of 6 subsets; in the remaining one the ordering reverses.

The margin is, however, **4.1× the seed standard deviation** (0.35 µg/m³), so it is not an artefact of random initialisation — it is fold-to-fold heterogeneity, not run-to-run noise. Restricting to the five reference-grade cities and excluding the low-cost Khujand fold, the model leads more clearly (25.84 against 28.51 µg/m³).

The defensible claim is therefore "lowest RMSE among admissible methods on this fold set", not "better than spatial interpolation". With 6 cities the benchmark cannot separate those two statements, and Section 6.2b makes the same point about the CAMS comparison. Full robustness table: `t7_05_ranking_robustness.csv`.

For reference, a constant equal to *the held-out city's own test-block mean* scores 28.12 µg/m³. **That predictor is not legal and is not a baseline.** Under leave-city-out the held-out city contributes no training label anywhere in the record, so its mean cannot be known at prediction time; it is reported as a diagnostic floor — the share of error that is pure within-city day-to-day variance — and the model is within 0.11 µg/m³ of it. An earlier version of this manuscript compared the model against that oracle and reported it as losing to "a constant". That comparison was invalid, and the correction is stated here rather than quietly dropped.

Three corrections to this comparison were needed:

1. Earlier drafts placed daily model scores beside baselines scored on *hourly* observations. Averaging removes within-day variance, so hourly RMSE is structurally larger; the apparent margin was arithmetic, not skill.
2. The comparison predated benchmark v1.1.0, in which the two Dushanbe records were found to be one instrument and merged (Section 2). Every fold's RMSE rose once the duplicate was gone.
3. The constant used was an oracle, as above.

6.1b The baseline ladder at a single resolution

Every rung below is scored on **the same daily evaluation rows as the learned model** (local-calendar daily means, ≥18 hours, the frozen 2024 test block). Table 6.1 above and the hourly ladder in Section 3 are not comparable to each other and are no longer presented as one ladder.

Model (daily, leave-city-out)	RMSE μg/m³
nearest_monitor	33.50
training_pool_mean	32.75
train_global_mean	32.70
train_global_median	30.99
ordinary_kriging	29.75
idw_k5_p2	29.44
LightGBM, retrospective (log target)	**28.01**

6.1c Per-fold R^2, including the negative folds

Section 3.6 requires per-city reporting from every submission to this benchmark. The same rule is applied here.

Held-out city	R^2 (vs that city's own mean)
Bishkek	-0.549
Dushanbe	-0.341
Ashgabat	-0.247

Khujand	+0.128
Tashkent	+0.323
Almaty	+0.517

3 of 6 folds are negative: in those cities the model is worse than predicting the city's own average. Averaging them into a single figure conceals that, which is why both the spread and the mean are reported.

6.2 Per-city results and the Diebold–Mariano tests

Pooling six cities into one number hides the finding.

Held-out city	LightGBM RMSE	CAMS RMSE	DM statistic	*p*
Almaty	15.30 ± 1.31	23.25	-4.06	0.0001
Ashgabat	20.74 ± 0.20	18.83	1.17	0.2420
Bishkek	20.21 ± 0.39	20.03	0.15	0.8805
Dushanbe	46.94 ± 0.79	47.96	-2.65	0.0085
Khujand (zero-shot)	38.81 ± 0.12	39.90	-2.03	0.0430
Tashkent	25.46 ± 0.60	28.63	-2.82	0.0050
pooled (n = 2214)	**31.49**	**33.07**	**-4.38**	**< 0.0001**

Tests use Newey–West HAC variance with the Harvey–Leybourne–Newbold small-sample correction. Negative statistics favour the learned model. One limitation of the test is stated here rather than left for a reader to raise: Diebold (2015) notes that the DM procedure was constructed to compare *given* forecasts, and that applying it to *estimated* models is its most common misuse. Both comparators here are estimated on training data, so the per-fold DM statistics below are read as descriptive diagnostics. The paper's inferential claim is the city-level test in Section 6.2b, not these. Six folds are tested, so per-fold *p*-values are additionally reported with a Holm step-down correction (Holm, 1979) in `t6_07_per_fold_holm.csv`; **3 of 6 survive it at α = 0.05.** The correction family is the 6 per-fold comparisons of the same model pair, declared here because a family chosen after seeing the *p*-values is not a correction.

6.2b Which *p*-value is the paper's claim

The pooled row above treats 2214 station-days from 6 cities as independent observations. They are not, in either dimension: the loss differential has first-order autocorrelation 0.25 within station, and station-days cluster within cities that contribute very unequal row counts. **That pooled figure is reported for continuity with the per-fold rows above and is not the paper's inferential claim.**

The estimand is the reduction in squared error at *a city with no local training labels*, so the unit of generalisation is the **city** — as Section 5.4 already argues, and as the leave-city-out protocol implies. The primary analysis therefore aggregates to one value per city and tests those 6 numbers.

	Test	Unit	*n*	*p*
Primary	paired *t* on city means	city	6	**0.1392**
Primary	exact sign-flip permutation	city	6	**0.1250**
Sensitivity	station-day, independence assumed	station-day	2214	2.6e-10
Sensitivity	station-day, Newey–West HAC (lag 60 d)	station-day	2214	0.0044
Sensitivity	cluster bootstrap over cities	city	6	0.0428

With 6 clusters, cluster-robust asymptotics are unreliable — the cluster-robust variance estimator is materially downward-biased below roughly 30–50 clusters (Cameron and Miller, 2015). Two remedies appropriate at this cluster count are reported together: a *t*-test on 6 city means with 5 degrees of freedom, and an exact sign-flip permutation test that assumes no distribution at all. The permutation test's smallest attainable two-sided *p*-value is 0.03125, a floor imposed by having only 6 cities; we state it rather than let a reader mistake it for evidence.

The evidence is suggestive and does not reach conventional significance under the unit of generalisation this benchmark is built around. Corrections for serial dependence alone leave the station-day result significant; treating cities as the unit does not. We report both and let the divergence stand, because it is a real property of a study with six cities, not a defect to be resolved by choosing the smaller number.

The pooled improvement is significant (< 0.0001); 4 of 6 individual folds are, and 3 of those survive Holm correction. Almaty, Tashkent (p = 0.0050) and Dushanbe hold after correction. Khujand (p = 0.0430) is significant uncorrected and is not after it, and we report it on that footing below.

In Ashgabat and Bishkek the sign is reversed and CAMS returns the lower RMSE. At Ashgabat that is 18.83 against 20.74 µg/m³ (DM statistic 1.17, p = 0.2420); at Bishkek the gap is smaller still and the test cannot separate it from zero (p = 0.8805). Neither reversal approaches significance, so the two methods are indistinguishable in those two cities rather than CAMS winning them.

Khujand, the zero-shot fold, clears significance before Holm correction (0.0430) and not after it at 38.81 µg/m³ against CAMS at 39.90. A city with no training rows anywhere in the record is predicted at least as well as the operational chemistry-transport model predicts it, which is the finding worth having. We stop short of calling it an improvement, because the correction removes it. It remains among the hardest folds in absolute terms, second only to Dushanbe.

Truncation-lag sensitivity. The daily comparisons in Table 6.2 use the automatic Newey–West bandwidth, floor(4(n/100)^(2/9)), which at n = 2214 is 7 days. An earlier hourly implementation instead derived the lag from the forecast horizon and passed `horizon_hours = 1` for the nowcasting task, which disables the HAC correction entirely and inflated *p*-values by roughly seven orders of magnitude. We now report the full sensitivity sweep: across truncation lags of 0 to 60 hours the pooled conclusion is stable, with all comparisons significant (yes) and a worst-case *p* of 0.0056.

Figure 3 gives the per-city comparison against debiased CAMS, and **Figure 5** the observed-versus-predicted scatter with the 1:1 line. The compression toward the mean visible in Figure 5 is the same effect that drives the RMSE/exceedance divergence of Section 4.3.

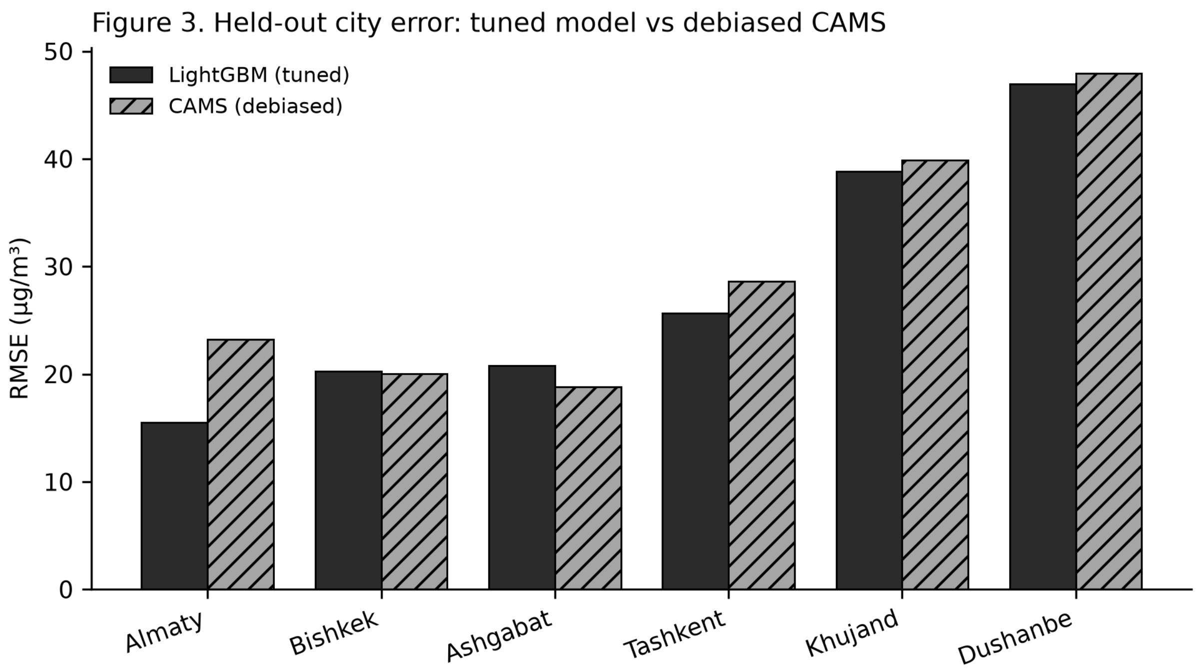

Figure 3. Held-out city RMSE (μg/m³), tuned LightGBM against debiased CAMS. Ashgabat is the fold where the sign reverses; the two are statistically indistinguishable there.

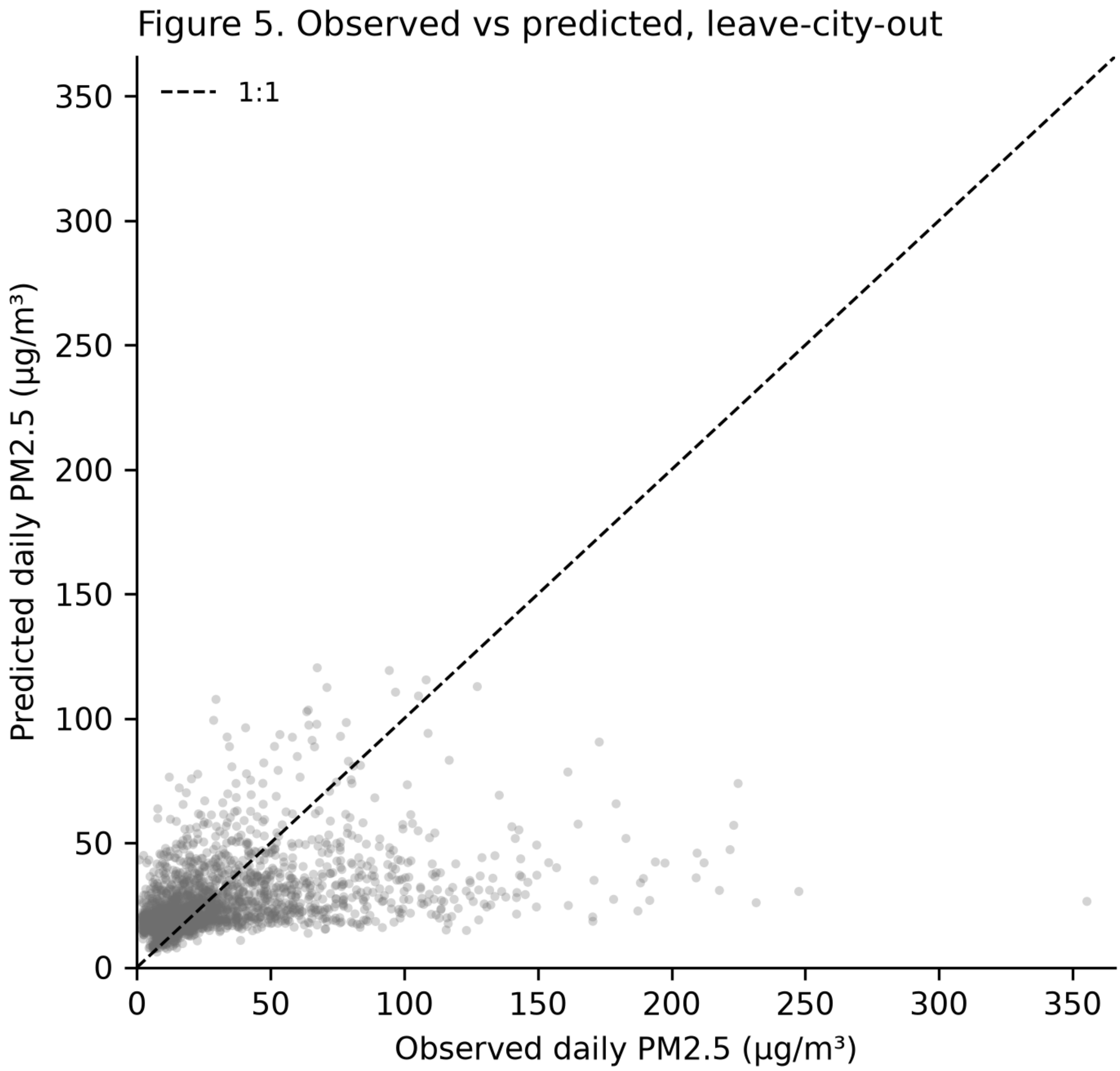

Figure 5. Observed against predicted daily mean PM2.5 (µg/m³) under leave-city-out, with the 1:1 line. Compression toward the mean is visible at high observed values and is the mechanism behind the RMSE/exceedance divergence of Section 4.3.

6.3 Task F — forecasting at monitored stations

What Task F is, precisely. The learned Task F model predicts the **daily** mean at a monitored station from that station's own history at lags of 1, 2 and 7 days plus 7- and 30-day rolling means. Its shortest lag is one day, so it is a **single-horizon, next-day (24 h) forecast evaluated at daily resolution**.

It is therefore **not comparable to the Task F baseline ladder in Section 3**, which is resolved across three horizons (24, 48 and 72 h) and scored on hourly observations (115,381 observations, against 2,214 daily rows for the model). Earlier drafts placed the two in one table. They measure different things at different resolutions, and the baseline row has been removed rather than rescaled: no honest rescaling exists, because the model does not produce 48 h or 72 h forecasts at all. Extending it to the full horizon set is left as future work and is not claimed here.

Feature set (daily, next-day horizon)	RMSE (µg/m³)	R^2
LightGBM, static only	22.03	0.58
LightGBM, deployable	21.58	0.59
LightGBM, retrospective	21.54	0.60

Task F is the easier problem and the numbers say so: R^2 = 0.60 against -0.04 for Task N. **These two figures must never be quoted together as though they described one system.** The Task F model may read the station's own recent history; the Task N model is predicting a city it has never seen. Reporting 21.54 µg/m³ as "the model's accuracy" would describe a capability the deployment does not have at unmonitored locations, which is the case the artefact exists to serve.

6.4 SHAP attribution: what actually carries the model

Attribution uses SHAP values (Lundberg and Lee, 2017), computed on the tuned model over the held-out folds.

Mean absolute SHAP over the test block, by feature family:

Family	Share of total attribution
spatial neighbour	**20.4%**
static geography	21.8%
calendar	22.2%
satellite	26.6%
CAMS forecast	9.0%

No feature family dominates. The five satellite products carry the largest share of attribution at 26.6%, ahead of calendar terms (22.2%), static geography (21.8%) and spatial neighbours (20.4%). The single largest individual feature is `doy_cos` at 0.18 mean absolute SHAP, a calendar term, ahead of `nbr_idw` at 0.12, the inverse-distance weighted neighbour concentration.

The spread across the top four families is narrower than the ranking suggests, and it is not stable enough to carry an interpretation on its own. Section 7.4 sets out why: on benchmark v1.0.0 this ordering came out the other way round, with spatial interpolation apparently ahead of the satellite record, and that ordering was an artefact of one duplicated station. Merging it moved every family. An attribution ranking computed on 7 instruments should be read as provisional, and the claim this paper previously drew from it is retracted in Section 7.4 rather than restated here.

Two further readings follow. First, **satellite retrieval-count features no longer appear at all**: an ablation on the validation block (Section 5.3) found them city-specific rather than transferable, and they are excluded from Task N. They remain in Task F, where the held-out entity is a time block rather than a city. Whether a retrieval failed still carries real information, which vindicates modelling missingness rather than dropping it (Section 2.4) and simultaneously indicates how little the retrieved values themselves add. Second, calendar features at 22.2% confirm that a large part of the signal is seasonal regularity that any climatology captures — consistent with the same-hour 7-day mean being the only Task F baseline with positive R^2.

Figure 4 shows attribution by feature family.

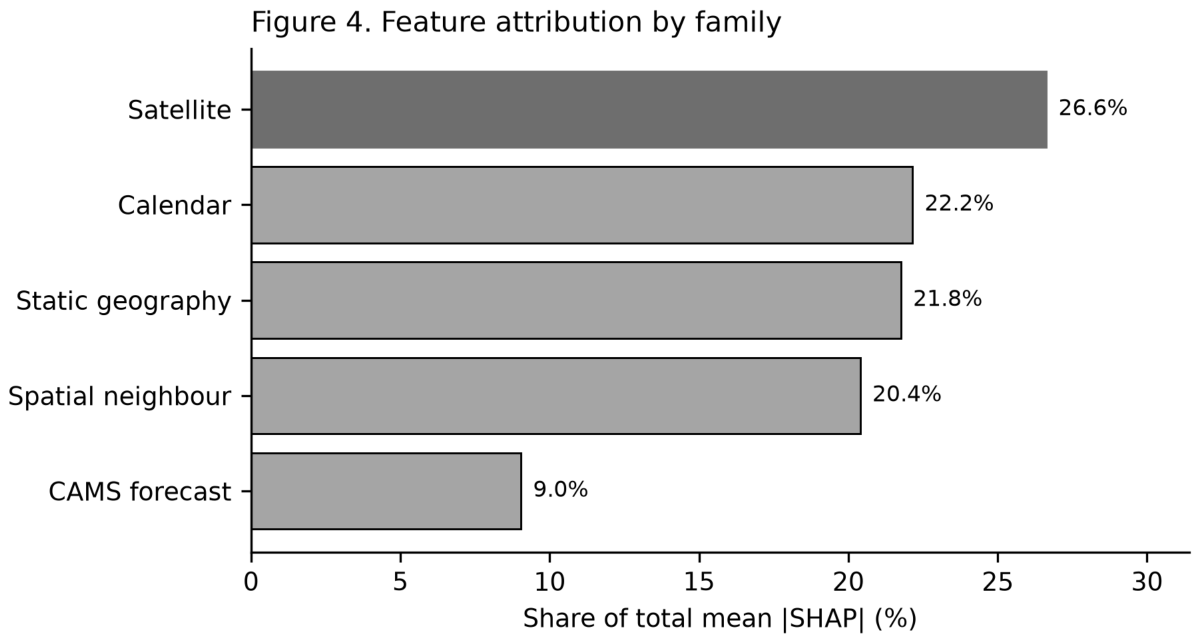

Figure 4. Feature attribution by family, as a share of total mean |SHAP|. Satellite families are hatched. On benchmark v1.1.0 the five satellite products together carry the largest share (26.6%), ahead of spatial neighbour features (20.4%) — a reversal of the pre-deduplication ordering discussed in Section 7.4.

6.5 Summary of what is and is not established

Established:

- **The benchmark itself**: 7 instruments across 6 cities, splits frozen and hash-verified, every reported number regenerated by one command.
- Deployability costs almost nothing: 28.56 vs 28.01 μg/m³. Restricting to features available at inference time is close to free, which is the one comfortable result here.

- The task is hard in a way the protocol makes visible. Under leave-city-out, a tuned gradient-boosting model with satellite, meteorological and spatial features does **not** improve on each city's own mean.

Not established — and previously claimed:

- **That the learned model beats pooled-debiased CAMS.** Under the city-level analysis this protocol implies, it does not: paired *t* $p = 0.1392$, exact permutation $p = 0.1250$. Only 3 of 6 folds survive Holm correction. The pooled station-day *p*-value reported in earlier drafts assumed an independence the data does not have.
- **That leading the ladder means the model is *useful* at unmonitored locations.** It has the lowest RMSE of any admissible method (28.01 μg/m³), but mean per-fold $R^2 = -0.04$, spread -0.55 to 0.52, with 3 of 6 folds negative. Lowest error and demonstrated within-city skill are different claims; only the first is established.
- **That spatial interpolation rather than satellite data drives the result.** That ordering reversed once the duplicated Dushanbe instrument was merged (Section 7.4). Satellite products now carry 26.6% against 20.4% for spatial neighbours. Neither ordering should be treated as robust: it inverted on the removal of a single station.

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7. Limitations and Discussion

This section is written for a reader hunting the study's weakest point. We would rather state it than have it found.

7.1 The labels themselves are provider-dependent in one city

The embassy monitors are published twice, by StateAir and by AirNow, under distinct identifiers. Ashgabat's pair is a clean duplicate: the two feeds agree to 0.1 μg/m³ on 100.0% of overlapping test-block hours. **Bishkek's pair is not.**

Across the full record the Bishkek feeds agree on 52.0% of overlapping hours, and the divergence concentrates in the frozen test year. Over 5389.00 overlapping hours in 2024 they agree on only **11.1%**, with a 95th-percentile disagreement of **33.6 μg/m³**. That is comparable to the RMSE of every model in this paper.

The implication is uncomfortable and irreducible. For Bishkek in the test block there is no single ground truth. Choosing the other provider's feed would shift the reported error by roughly the margin that separates our models from each other. Bishkek's DM result ($p = 0.8805$, not significant) should be read with that in view, and we claim no improvement there. Merging the feeds, which we do, reduces variance. It cannot manufacture a label the two publishers agree on. **This limits the data source, not the pipeline, and no modelling choice removes it.**

7.2 Leave-station-out covers a minority of the benchmark

2 leave-station-out folds exist and they span two cities. **Almaty, Ashgabat, Bishkek, Dushanbe, Tashkent each hold a single instrument**, so within-city station holdout is undefined there.

The consequence runs deeper than reduced coverage. The Q6 timezone check compares instruments within a city. In a single-station city, a constant lifelong offset is invisible to every rule in the suite, and the QC output records that explicitly instead of returning a pass that actually means "not tested". Four of six cities therefore rest on metadata correctness for their time alignment. We regard this as the benchmark's largest unaudited assumption.

A related constraint sits upstream: 306 candidate stations were excluded for insufficient span, almost all of them low-cost units whose measurement uncertainty is well documented (Zheng et al., 2018; Crilley et al., 2018).

That exclusion was not applied uniformly, and the exception matters. **Khujand's two instruments are Clarity low-cost sensors** (`is_monitor = false` in the OpenAQ census), not reference-grade monitors. Every other city in the benchmark is a US-embassy BAM/FEM-class monitor published by AirNow or StateAir. Khujand is therefore the one city whose labels carry the measurement uncertainty the paragraph above cites, and it is also the city that contributes no training rows -- so its fold reports low-cost labels scored against a model that never saw them.

The mechanism matters for where this benchmark sits. Optical sensors size particles by light scattering, so their response depends on humidity and on aerosol composition: Crilley et al. (2018) attribute the dominant positive bias in this sensor class to hygroscopic growth, and Barkjohn et al. (2021) show that removing the resulting bias needs a correction fitted against reference monitors across a network. Neither condition is met here. The published corrections are derived in humid, sulphate- and organic-dominated environments, and this region is arid and dust-dominated, so their coefficients do not transfer; and Khujand has no co-located reference instrument to fit a local correction against. We therefore report the Clarity readings as published, label the fold, and treat its errors as carrying an instrument-uncertainty component we cannot separate from model error.

Two consequences follow. First, results for the Khujand fold are not comparable in kind to the other five and should not be read as a reference-grade generalisation test. Second, the pre-registered 2-year span rule is satisfied for these two stations only by counting observations after the benchmark record ends: inside the window their spans are **1.09 y and 1.07 y**, against a stated minimum of 2 y. Admitting them was a coverage decision, and it is recorded here as one.

Leave-city-out stays the primary spatial protocol for one reason: it is available for all 6 cities. Leave-station-out is reported where possible and supports no headline claim.

7.3 The model does not beat CAMS everywhere

4 of 6 leave-city-out folds reach significance before correction and 3 after it. In **Ashgabat and Bishkek the sign is reversed**: at Ashgabat, LightGBM returns 20.74 µg/m³ against CAMS at 18.83 µg/m³, DM statistic 1.17, p = 0.2420. Read correctly, the two are indistinguishable there. CAMS does not win. Across the fold set, the pooled result (< 0.0001) rests on Almaty, Tashkent (p = 0.0050) and Dushanbe, the three folds that survive Holm correction. Khujand is significant uncorrected and not after correction. Bishkek (p = 0.8805) returns a marginally higher RMSE than CAMS, not a lower one, and the difference is indistinguishable from zero either way.

Khujand is worth pausing on even though the correction removes it, since it is the fold with no training label at all. Zero-shot transfer into an unmonitored city is not where a reader would expect the method to hold up best, and it does not collapse there: the fold is significant before Holm and the model's RMSE sits below the chemistry-transport model's. We draw the weaker of the two available conclusions from that. It is evidence that the spatial machinery is not simply memorising the training cities, and it is not evidence of an improvement, because a correction the paper itself imposes takes the improvement away.

Ashgabat is the opposite case, and the one where the model has least to work with. Turkmenistan operates no national monitoring network, so the fold reduces to a single embassy instrument with no domestic context, and its nearest benchmark neighbour sits far outside any plausible interpolation radius. Under exactly those conditions a chemistry-transport model with a physical emissions inventory is a strong comparator. It should be.

7.4 What carries the model, and how that changed

Section 6.4 reports mean absolute SHAP by feature family. On benchmark v1.1.0 the five satellite products together account for **26.6%** of attribution, spatial neighbour features for 20.4%, and static geography a further 21.8%. The top single feature is `doy_cos`.

This reverses the ordering reported in earlier drafts of this work, and the reason is instructive rather than incidental. Under benchmark v1.0.0 the two Dushanbe records were treated as separate stations when they are one instrument republished twice (Section 2, D-012). Every other city therefore had an additional neighbour at effectively zero distance from an existing one, which inflated the apparent value of spatial interpolation. Once the duplicate is merged, satellite attribution overtakes it.

The earlier claim — that a study assembled around remote sensing was "really" a spatial interpolator — was an artefact of a duplicated station. We state that plainly because it was published as a finding reported against interest, and it is no longer supported. Three qualifications apply to the current ordering.

1. *This is a property of the protocol as much as of the products.* Leave-city-out asks for a concentration where no monitor exists. Neighbour information is the most direct route to that answer, and satellite columns are a weak proxy for surface concentration under any protocol whatsoever.

2. *Missingness is informative but does not transfer between cities.* Retrieval-count features were promoted to predictors on the evidence that missingness is target-correlated. A validation-block ablation (Section 5.3) then showed they **hurt** leave-city-out generalisation, because retrieval success depends on local surface brightness, snow cover and solar geometry — properties of a particular city. They are excluded from Task N and retained for Task F. That is a finding about retrieval physics, and at the same time a measure of how little the retrieved values contribute.

3. *SO_2 is structurally absent in the season it exists to observe.* Retrieval needs ultraviolet signal; at these latitudes in December it falls to 0.1% against 91.0% in July. The direct tracer for the region's dominant winter source is unavailable throughout that source's season, so its low attribution is partly a measurement-geometry artefact, not evidence that SO_2 carries no information.

We report the attribution as measured. A paper arguing that satellite remote sensing enables air quality prediction in Central Asia would not be supported by these experiments.

7.5 Zero-drift reporting, and the drift it caught

Every number in this manuscript is a double-brace placeholder token resolved at render time from `paper/tables/*.csv`. An unresolved placeholder is a hard build failure. A test also checks a sample of manuscript figures directly against the CSVs, bypassing the intermediate `numbers.json` entirely, so that a bug in the extractor cannot certify itself.

The mechanism earned its place during drafting. Section 2's missingness statistics were originally transcribed from a console output. When they were finally banked to a table, the transcribed values proved wrong: SO_2 retrieval had been written as 61.5% against a recomputed 57.4%, and four further figures were off in the same direction. The errors were small and none of them changed a conclusion, which is exactly why no reader would ever have caught them. Regenerating the table also exposed that the two Section 2 tables had no producer script at all and could not be rebuilt by `make reproduce`. Both have one now.

7.6 Scope of the record

- **The record ends before the source does.** Five of the ten contributing source feeds stop on 2025-03-04, when the StateAir publication channel closed, and at benchmark-station level **2 of 7 stations (8881, Bishkek) end there**; the rest survive through a longer-lived feed. **No result in this paper speaks to current conditions.** The monitors did not all stop, though: as of 2026-08-14 the same diplomatic-post instruments are still republished through AirNow at Ashgabat (to 2025-09-24), Almaty (to 2025-11-14) and Dushanbe (still reporting). The record *can* therefore be extended for those cities, but doing so would change the benchmark and requires a version bump, not a silent refresh.
- **Kazakhstan contributes one city.** Astana failed the completeness rule at 42.8% against a required 60%. The largest country in the region is represented by Almaty

alone.

- **Six cities is a small spatial sample.** Fold-to-fold standard deviation (11.38 μg/m³) exceeds seed standard deviation (1.79 μg/m³) by roughly an order of magnitude. Conclusions are far more sensitive to which cities are in the set than to any training randomness.
- **ERA5 is oracle-only and incompletely retrieved.** Its measured latency (163 h) exceeds every evaluated horizon, so it cannot enter the deployable set. The multi-year retrieval was stopped once that was established.
- **One item of prior art is unresolved.** A 2025 conference abstract claims the first machine-learning PM2.5 prediction for Tashkent. The publisher page returns HTTP 403 and the record is absent from OpenAlex, Crossref, Semantic Scholar and Europe PMC, so **its split protocol could not be verified**. Section 1.4 states what we consequently do not claim. Should that work prove spatially stratified, the contribution of C1 narrows from "first such protocol in the region" to "first multi-city benchmark in the region". The benchmark is unaffected either way. Only the framing of its novelty moves.
- **Literature depth is uneven, and recorded as such.** Of 30 sources, 16 were read in full and 6 at abstract depth; 7 remain at snippet depth behind publisher paywalls. For those, bibliographic identity is verified against Crossref/OpenAlex but *content* claims are not independently confirmed. `research/LITERATURE.md` records the two axes separately rather than averaging them into a single confidence.
- **Russian-language coverage is incomplete.** CyberLeninka and eLIBRARY.RU are not indexed by the APIs used here, so the falsifier F4 verdict, that no equivalent regional benchmark exists in the Russian-language literature, remains provisional.

7.7 Train/serve skew is measured but not eliminated

The deployable set uses near-real-time satellite products while the benchmark is built on standard-latency ones. These are different processing streams over the same instrument, and we have not quantified the distributional difference between them. A service trained on standard products and served NRT products is exposed to skew that this study bounds only indirectly, through the small deployable/retrospective gap of Section 5.5. That gap is evidence that latency-restricted *feature sets* cost little. It is not evidence that NRT and standard versions of the same product are interchangeable. Quantifying the difference is the most immediate piece of follow-up work, and it needs a parallel NRT archive we did not have.

7.8 What the benchmark is for

The headline modelling result is modest. R^2 = -0.04 at unmonitored locations, no improvement over CAMS that separates from zero once cities rather than station-days are treated as the unit of generalisation (paired *t* *p* = 0.1392; exact permutation *p* = 0.1250), and an attribution profile in which no feature family dominates. We consider that the appropriate outcome.

The contribution is the fixed evaluation. Before this benchmark existed, a Central Asian air quality result could be reported on a random split, with reanalysis features unavailable at inference time, against no baseline ladder, scored with an exceedance F1 that a constant already achieves (0.741 at a 61.8% base rate, because the region's air is bad on most days, not because the classifier is good). Every one of those choices would have produced a more impressive paper than this one. The splits are frozen and checksummed. The protocol violations are enforced by failing tests rather than requested in prose. The numbers above are what survives that. A future model that genuinely improves on 28.01 µg/m³ under this protocol will have demonstrated something real.

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8. Conclusion

We built the benchmark this region did not have. 7 instruments across 6 cities (5 reference-grade, 2 low-cost), splits frozen and checksummed before the reported results were produced, two tasks kept separate, and a baseline ladder that every future submission has to climb. The splits are immutable by test, and that test fails for us exactly as it fails for anyone else.

The modelling result is modest and is reported without inflation. Tuned gradient boosting reaches RMSE 28.01 ± 0.35 µg/m³ at unmonitored locations, the lowest of any admissible baseline including inverse-distance weighting (29.44 µg/m³) — but that ordering is not statistically separable (paired p = 0.586) and is not robust to removing a single city. Its advantage over bias-corrected CAMS is likewise **not statistically significant** when the city is the unit of analysis (p = 0.1392), and mean per-fold R^2 is -0.04 with 3 of 6 folds negative. Lowest error and demonstrated skill are not the same claim, and only the first is supported here. Every protocol choice available to us would have produced a larger one: a random split, reanalysis features unavailable at inference, no baseline ladder, an exceedance F1 that a constant classifier already achieves. The number reported here is what survives after those escapes are closed by failing tests.

Three findings run against the study's own framing. A trivial always-exceed classifier is not beaten by any credential-free nowcaster, because these cities exceed the WHO 24-hour guideline on most days — a fact about the region, not about the models. Attribution is spread almost evenly across feature families, with the five satellite products (26.6%) narrowly ahead of spatial interpolation (20.4%) rather than displaced by it. That ordering is the reverse of the one reported before the duplicate Dushanbe instrument was found and merged, and the margin between the two is small enough that we treat the ranking as provisional (Section 7.4). And measured acquisition latency invalidated three of five initial availability assumptions, one of them by a factor of roughly 4,600.

For practitioners. The clearest practical signal in this work is not an ordering of feature families — that ordering proved fragile, reversing once a single duplicated instrument was removed. It is that with 7 instruments across 6 cities, a learned model does not beat knowing a city's own average, and the attribution ranking is unstable to one station. Both point the same way: the binding constraint is network density, not modelling technique or choice of remote-sensing product. The single most valuable investment for Central Asian air quality modelling is more stations — particularly in Turkmenistan, which has none, and Kazakhstan, which contributes one city here.

For the field. The benchmark's value is that it forecloses the shortcuts. A future model that genuinely improves on 28.01 µg/m³ under this protocol will have demonstrated something real, and the comparison will be like-for-like because the splits cannot move. Priorities for extension, in order: additional cities to reduce the fold-to-fold variance that presently exceeds seed variance by an order of magnitude; a parallel near-real-time satellite archive to quantify train/serve skew; and a post-2025 ground-truth source to replace the publication channel that closed on 2025-03-04.

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9. Declarations

9.1 Data availability

The benchmark definition — station set, temporal blocks, leave-city-out and leave-station-out folds — is archived in Zenodo at **https://doi.org/10.5281/zenodo.21930669** (version 1.1.0, CC BY 4.0), together with the SHA-256 checksum that fixes it. The working repository is `https://github.com/jalencik/eco-pulse-benchmark`; cite the Zenodo version DOI rather than the branch head, so a reported score is attributable to one frozen split definition. All analysis code, the baseline ladder, and the scripts that regenerate every table and figure in this manuscript are in the same repository. A single command (`make reproduce`) regenerates every reported number from the frozen splits.

Ground observations derive from the OpenAQ API (US diplomatic-post reference monitors and national networks). The derived hourly panel is **not** redistributed here because the per-station licence terms have not been transcribed; `data/MANIFEST.md` records provenance per source, and `README.md` gives the two commands that rebuild the panel from the API with an OpenAQ key. Satellite products are public: MAIAC AOD (MCD19A2.061) and Sentinel-5P (OFFL/NRTI) via Google Earth Engine; CAMS forecasts and ERA5 via the Copernicus Atmosphere Data Store and Climate Data Store respectively.

Note on permanence. The US State Department terminated its global diplomatic-post air quality programme in March 2025. Five of the ten contributing source feeds stop on 2025-03-04 — every StateAir feed, plus Bishkek's AirNow feed — and at benchmark-station level, after co-published feeds are merged, **2 of 7 stations (8881, Bishkek) end there**; the others survive through their longer-lived feed. An earlier version of this statement said "six of the eight", which was wrong in both terms. The record curated here is finite and partly withdrawn at source, which is a reason to archive this benchmark rather than an argument against it. Authors intending to cite it should reference the frozen checksum, not the branch head.

9.2 Declaration of generative AI in the writing process

During the preparation of this work the corresponding author, Jaloliddin Musayev, used Anthropic Claude (Claude Code) to assist with software implementation, data-pipeline construction, statistical tooling, and drafting and editing of the manuscript text. The authors reviewed and edited all output, verified every reported number against the regenerated result tables, and take full responsibility for the content of the publication. Generative AI is not listed as an author and no AI system holds authorship or accountability for this work.

Note for submission: adjust this wording to the target journal's template. Elsevier requires the statement immediately above the reference list. IEEE additionally requires the system to be named, the affected sections identified, and the level of use described. Routine grammar and spelling correction does not, under any of these policies, require declaration.

9.3 CRediT author contributions

Jaloliddin Musayev: Conceptualisation; Methodology; Software; Validation; Formal analysis; Investigation; Data curation; Writing — original draft; Writing — review and editing; Visualisation; Project administration.

Asadbek Abdivayitov: Data curation; Investigation.

Ozodbek Yo'ldashev: Supervision; Writing — review and editing.

All authors read and approved the submitted manuscript.

9.4 Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. All computation was performed on the authors' personal hardware, and every data source used is publicly accessible at no cost.

9.5 Competing interests

The authors declare no competing financial or non-financial interests.

9.6 Ethics

This study used publicly available environmental measurements and involved no human subjects, animal subjects, or personally identifiable data. Ethical approval was not required.

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Sought but not obtained

Listed so the gap is visible rather than absent. No claim in this manuscript rests on these.

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